The documentation and process conversion measures necessary to comply with this document shall be completed by 15 July 2020.

INCH-POUND

MIL-PRF-19500/590M
15 March 2020
SUPERSEDING
MIL-PRF-19500/590L
12 July 2017

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, ULTRAFAST RECOVERY, POWER RECTIFIER,
DEVICE TYPES 1N6626 THROUGH 1N6631,
QUALITY LEVELS JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for a silicon, ultrafast recovery, semiconductor power rectifier diode. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each device type as specified in MIL-PRF-19500.

1.2 Package outlines. The device packages for the encapsulated device types are as follows: axial through-hole mount similar to DO-204AL (formerly DO-41) in accordance with figure 1 and surface mount version in accordance with figure 2.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

1.3.1 Ratings applicable to all types. Ratings applicable to all part or identifying numbers (PIN). T_{STG} = −65°C to +175°C. T_J = +150°C maximum.

1.3.2 Ratings applicable to individual types.

Col. 1	Col. 2	Col. 3	Col. 4	Col. 5	Col 6	Col. 7	Col. 8	Col. 9	Col. 10
Types (1)	V_{RWM}	I_{OL} at T_L = +75°C (2) (3) (4)	I_{O2} at T_A= +25°C (2) (5) (6)	I_{FSM} at t_p= 8.3 ms	Baro-metric pressure	t_{rr} (7)	$R_{\theta JL}$ (8)	$R_{\theta JEC}$	$R_{\theta JX}$
	V dc	A	A	A(pk)	mm Hg	ns	°C/W	°C/W	°C/W
1N6626	200	2.3	1.75	75	8	30	22	6.5	50
1N6627	400	2.3	1.75	75	8	30	22	6.5	50
1N6628	600	2.3	1.75	75	8	30	22	6.5	50
1N6629	800	1.8	1.40	75	33	50	22	6.5	50
1N6630	900	1.8	1.40	75	33	50	22	6.5	50
1N6631	1,000	1.8	1.40	60	33	60	22	6.5	50

See notes on next page.

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1.3.2 Maximum ratings − Continued.

(1) Includes both axial leaded package types (non-surface mount) and surface mount (US suffix) types.
(2) Average current with a half-sine wave including reverse voltage amplitude equal to the magnitude of the full rated V_{RWM}.
(3) Derate linearly 1.33 percent/°C for T_L > +75°C.
(4) These rated currents also apply to U or US suffix types when the maximum temperature of the end-caps (mounting surface) is +110°C; derate linearly 2.5 percent/°C above T_{EC} > +110°C.
(5) Derate linearly 0.80 percent/°C for T_A > +25°C.
(6) The 1.4 amp rating at +25°C ambient is for thermal (PC boards or other) mounting methods where thermal resistance from mounting point to ambient is still sufficiently controlled where $T_{J(MAX)}$ in 1.3.1 is not exceeded. This equates to $R_{\theta JX} \leq 50$°C/W as shown. Also see application notes in 6.7.1 for the worst-case 1N6631.
(7) The reverse recovery time (method 4031 of MIL-STD-750, condition B) at T_J = +125°C shall not exceed three times the +25°C limit. Exceeding T_J = +125°C may change reverse recovery times at +25°C to higher levels as indicated in accelerated life testing in 4.4.2.1 B5 for JANS or other life testing in 4.4.2.2 B3 and 4.4.3.1 C6.
(8) Measured at L = .375 inch (9.52 mm). See figures 3 and 4 for thermal impedance curves.

1.4 Primary electrical characteristics. Unless otherwise noted, T_A = + 25°C.

Col. 1	Col. 2	Col. 3	Col. 4	Col. 5	Col. 6		Col. 7	
Types (1)	I_{R1} at T_J = +25°C	I_{R2} at T_J = +150°C	$I_{RM(REC)}$ at 2 A, 100 A/µs	C_T at V_R = +10 V	V_{FM1} at I_{F1}		V_{FM2} at I_{F2}	
					V_{FM1}	I_{F1}	V_{FM2}	I_{F2}
	µA	µA	A pk	pF	V	A	V	A
1N6626	2.0	500	3.5	40	1.5	4.0	1.35	2.0
1N6627	2.0	500	3.5	40	1.5	4.0	1.35	2.0
1N6628	2.0	500	3.5	40	1.5	4.0	1.35	2.0
1N6629	2.0	500	4.2	40	1.7	3.0	1.40	1.4
1N6630	2.0	500	4.2	40	1.7	3.0	1.40	1.4
1N6631	4.0	600	5.0	40	1.95	2.0	1.60	1.4

(1) Includes both axial leaded package types (non-surface mount) and surface mount (US suffix) types.

1.5 PIN. The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example, 6.5 for a list of available PINs, and 6.6 for supersession information.

1.5.1 JAN certification mark and quality level.

1.5.1.1 Quality level designators for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", JANTXV, and "JANS".

1.5.2 Device type. The designation system for the device types covered by this specification sheet are as follows.

1.5.2.1 First number and first letter symbols. The diodes of this specification sheet use the first number and letter symbols "1N".

1.5.2.2 Second number symbols. The second number symbols for the diodes covered by this specification sheet are as follows: "6626", "6627", "6628", "6629", "6630" and "6631".

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1.5.3 Suffix symbols. The following suffix letters are incorporated in the PIN as applicable:

	A blank suffix symbol indicates a round body, axial through-hole mount (DO-204AL) package (see figure 1).
US	Indicates a round body, square endcap surface mount package (see figure 2).

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QML-19500.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

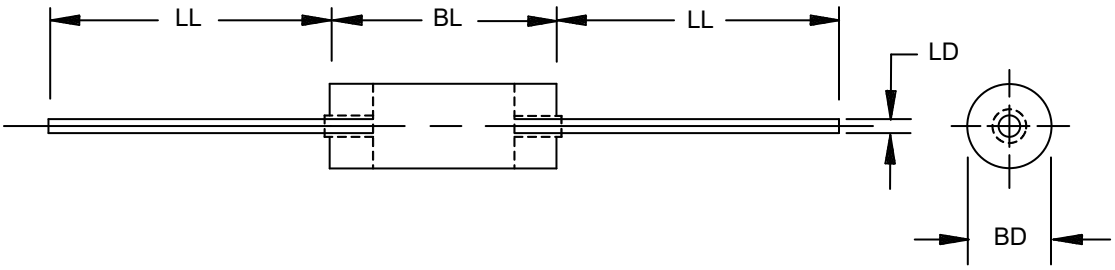
DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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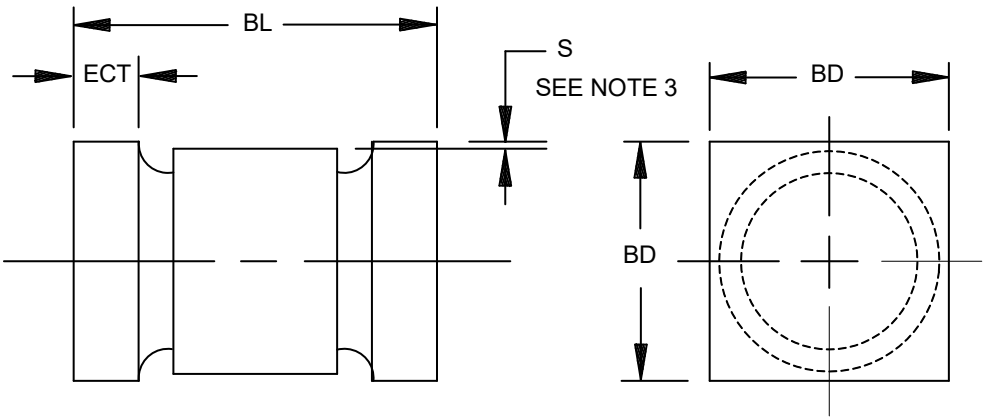


Symbol	Dimensions				
	Inches		Millimeters		
	Min	Max	Min	Max	Notes
BD	.115	.137	2.92	3.48	3
BL	.130	.300	3.30	7.62	4
LD	.037	.042	0.94	1.07	4
LL	.900	1.300	22.86	33.02	

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Dimensions are pre-solder dip.
3. Dimension BD shall be measured at the largest diameter.
4. The BL dimension shall include the entire body including slugs and sections of the lead over which the diameter is uncontrolled. This uncontrolled area is defined as the zone between the edge of the diode body and extending .050 inch (1.27 mm) onto the leads.
5. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions of round body, axial leaded package (similar to DO-204AL, formerly DO-41).

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Dimensions				
Symbol	1N6626US through 1N6631US			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.200	.225	5.08	5.72
BD	.137	.148	3.48	3.76
ECT	.019	.028	0.48	0.71
S	.003		0.08	

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Dimensions are pre-solder dip.
3. The "S" dimension is the minimum clearance of glass body to mounting surface on all orientations.
4. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 2. Physical dimension of round body, square end cap (US) surface mount package.

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows:

EC .....End-cap.
$I_{RM(REC)}$ .....Peak reverse recovery current.
P_{RM}.....Peak reverse power.
V_{FRM} .....Forward recovery voltage.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figures 1 (similar to DO-204AL) and 2 (surface mount) herein.

3.4.1 Lead finish. Unless otherwise specified, the lead or end cap finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. When solder alloy is used for finish the maximum lead temperature is limited to 175°C maximum. Where a choice of finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Diode construction. These devices shall be constructed utilizing non-cavity double plug construction with high temperature metallurgical bonding between both sides of the silicon die and terminal pins (see MIL-PRF-19500). Metallurgical bond shall be in accordance with the requirements of category I in MIL-PRF-19500. The US version devices shall be structurally identical to the non-surface mount devices except for lead terminations.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5.1 Marking for US devices. For US version devices only, all marking may be omitted from the device except for the cathode marking. All marking which is omitted from the body of the device shall appear on the label of the initial container. At the option of the manufacturer, US version devices may include laser marking on an end cap, to include part number and lot date code for all levels. JANS levels shall also include serialization. The prefixes JAN, JANTX, JANTXV, or JANS may be abbreviated as J, JX, JV, or JS, respectively. (Example: The part number may be reduced to JS6626). All device marking, except for polarity and serial numbers, shall also appear on the unit package used as the initial protection for delivery.

3.5.2 Polarity. The polarity of all types shall be indicated with a contrasting color band to denote the cathode end. Alternatively, for U and US suffix devices, a minimum of three contrasting color dots spaced around the periphery on the cathode end of the device may be used.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I, II, and III).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E qualification shall be performed herein for qualification or requalification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot to this revision to maintain qualification.

4.3 Screening (JANS, JANTX, and JANTXV levels). Screening shall be in accordance with appendix E, table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurement	
	JANS level	JANTXV and JANTX level
1a	Required	Not required
1b	Required	Required (JANTXV only)
2	Optional	Optional
3a	Required	Required
(1) 3c	Thermal impedance (see 4.3.1)	Thermal impedance (see 4.3.1)
4	Not applicable	Not applicable
5	Not applicable	Not applicable
6	Not applicable	Not applicable
7a	Not applicable	Not applicable
7b	Optional	Optional
8	Required	Not required
9	Required I_{R1} and V_{FM1}	Not required
10	Method 1038 of MIL-STD-750, condition A	Method 1038 of MIL-STD-750, condition A
11	I_{R1} and V_{FM1}, ΔI_{R1} = ±400 nA dc (±800 nA dc for 1N6631) or 100 percent of initial reading; whichever is greater. $\Delta V_{FM1} \leq \pm 0.05$ V dc	I_{R1} and V_{FM1}
12	Required, see 4.3.2	Required, see 4.3.2

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4.3 Screening (JANS, JANTX, and JANTXV levels) - Continued.

Screen	Measurement	
	JANS level	JANTXV and JANTX level
13	Subgroups 2 and 3 of table I herein: ΔI_{R1} = ±400 nA dc (±800 nA dc for 1N6631) or 100 percent of initial reading; whichever is greater; ΔV_{FM1} = ±0.05 V dc; scope display evaluation, method 4023 of MIL-STD-750, see 4.5.4	Subgroup 2 of table I herein: ΔI_{R1} = ±400 nA dc (±800 nA dc for 1N6631) or 100 percent of initial reading; whichever is greater; ΔV_{FM1} = ±0.05 V dc; scope display evaluation, method 4023 of MIL-STD-750, see 4.5.4
14a 14b	Not applicable Required	Not applicable Required
15	Required	Not required
16	Required	Not required

(1) Thermal impedance may be performed any time after temperature cycling is performed in accordance with MIL-PRF-19500, screen 3a; JANTX and JANTXV levels do not need to be repeated in screen 13.

4.3.1 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 or 4081 of MIL-STD-750 as applicable, using the guidelines in that method for determining I_M, I_H, t_H, t_{MD} and K factor where appropriate). See table II, subgroup 4 and figures 3 and 4 herein.

4.3.2 Free air power burn-in. Power burn-in shall be in accordance with condition B of test method 1038 of MIL-STD-750. The details of 4.5.3 and those following shall apply: T_A = +55°C maximum. Adjust I_O or T_A to achieve the required T_J. T_J = +115°C minimum and +150°C maximum. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, tables E-VIA (JANS) and E-VIB (JAN, JANTX and JANTXV) of MIL-PRF-19500 and 4.4.2.1 and 4.4.2.2 herein. Delta measurements shall be in accordance with table III herein.

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4.4.2.1 Quality level JANS. For B5, if a failure occurs, resubmission shall be at the test conditions of the original sample.

Subgroup	Method	Condition
B3	1056	0°C to +100°C, 25 cycles.
B3	1051	−55°C to +175°C, 100 cycles.
B3	4066	Condition A, I_{FSM} = rated I_{FSM} (see 1.3.2, col. 5); ten surges of 8.3 ms each at 1 minute intervals, I_O = 0, V_{RWM} = 0.
B4	1037	$I_O = I_{O2}$ rated minimum (see 1.3.2, col. 4); V_R = rated V_{RWM} (see 1.3.2, col. 2 and 4.5.3); 2,000 cycles.
B5	1027	$I_O = I_{O2}$ rated minimum (see 1.3.2, col. 4 and 4.5.3); adjust T_A and or I_O to achieve T_J = +150°C minimum; f = 50 to 60 Hz; n = 45, c = 0; t = 1,000 hours. T_A = +55°C maximum. Temporary leads may be added for surface mount devices. For irradiated devices, include t_{rr} as an end-point measurement. Delta t_{rr} shall not exceed 60 percent of initial reading. With approval of the qualifying activity, alternate life test criteria (hours, bias condition, T_J, mounting conditions) may be used.
B6	4081	See 4.3.1. For axial devices, $R_{\theta JL}$ (maximum) ≤ 22°C/W; L = .375 inch (9.53 mm). For US devices, $R_{\theta JEC}$ ≤ 6.5°C/W.
B8	4065	P_{RM} ≥ 636 W for square wave in accordance with method 4065 of MIL-STD-750 (P_{RM} ≥ 1000 W for half-sine wave). Test shall be performed on each sublot; sampling plan n = 10, c = 0, electrical end-points, see 4.4.2.

4.4.2.2 Quality levels JAN, JANTX, and JANTXV.

Subgroup	Method	Condition
B2	1056	0°C to +100°C, 10 cycles.
B2	1051	−55°C to +175°C, 25 cycles.
B3	1027	$I_O = I_{O2}$ (rated see 1.3.2, col. 4) minimum, adjust I_O to achieve the required T_J of +125°C minimum. Apply V_R = rated V_{RWM} (see 1.3.2, col. 2), f = 50 to 60 Hz (see 4.5.3.1). For irradiated devices, include t_{rr} as an end-point measurement. Delta t_{rr} shall not exceed 25 percent of initial reading.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VII of MIL-PRF-19500. See table III for delta limits when applicable.

Subgroup	Method	Condition
C2	1056	−0°C to +100°C, 10 cycles.
C2	1051	−55°C to +175°C, 25 cycles.
C2	2036	Axial devices − Tension: Test condition A; weight = 12 pounds (5.44 kg) for silver leads; 20 pounds (9.07 kg) for copper leads, t = 15 seconds. Lead fatigue: Test condition E; weight 2 pounds (0.9 kg). (Not applicable to US devices).
C2	2038	US devices − Tension: Test condition B; weight = 12 pounds (5.44 kg) for silver leads; 20 pounds (9.07 kg) for copper leads, t = 15 seconds.
C5	4081	See 4.3.1. For axial devices, $R_{\theta JL}$ (maximum) ≤ 22°C/W; L = .375 inch (9.53 mm). For US devices, $R_{\theta JEC}$ ≤ 6.5°C/W.
C6	1027	$I_O = I_{O2}$ (rated see 1.3.2, col. 4) minimum. adjust I_O to achieve the required T_J of +125°C minimum. Apply V_R = rated V_{RWM} (see 1.3.2, col. 2), f = 50 to 60 Hz (see 4.5.3.1). For irradiated devices, include t_{rr} as an end-point measurement. Delta t_{rr} shall not exceed 25 percent of initial reading.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-IX of MIL-PRF-19500 and as specified herein. See table III for delta limits when applicable.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Inspection conditions. Unless otherwise specified, all inspections shall be conducted at T_A = room ambient as defined in the general requirements of MIL-STD-750.

4.5.3 Burn-in and life tests. These tests shall be conducted with a half-sine waveform of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall be neither greater than 180 degrees, nor less than 150 degrees.

4.5.3.1 Burn-in. The use of a current limiting or ballast resistor is permitted provided that each DUT still sees the full I_O (minimum) and that the minimum applied voltage, where applicable, is maintained throughout the burn-in period.

4.5.3.2 Junction temperature measurements. Use method 3100 of MIL-STD-750 to measure T_J.

4.5.4 Scope display evaluation. Scope display evaluation shall be stable in accordance with condition A of method 4023 of MIL-STD-750. Scope display may be performed on ATE (automatic test equipment) for screening only with the approval of the qualifying activity. Scope display in table I, subgroup 4 shall be performed on a scope. Reverse current (I_{BR}) over the knee shall be 500 µA peak.

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4.5.5 Reverse recovery time.

4.5.5.1 Low current reverse recovery time. The low current reverse recovery time shall be measured in accordance with condition B of test method 4031 of MIL-STD-750. The following details shall apply: The test circuit of figure 5 or equivalent shall be used. The low current reverse recovery time shall be 0.5 A forward current to 1.0 A reverse current. The reverse recovery time is defined as the time the rectifier begins to conduct in the reverse direction (crosses I = 0) until the reverse current decays to −0.25 A. The point of contact on the leads shall be no less than .375 inch (9.52 mm) from the diode body.

4.5.5.2 High current reverse recovery time. The high current reverse recovery time shall be measured in accordance with condition D of test method 4031 of MIL-STD-750. The following details shall apply: The test circuit of figure 6 or equivalent shall be used. A pulse of forward current for the DUT is provided through S_1 and controlled by V_3, R_{L1} and the timing input pulse V_1. The duration of pulse, t_1, should be sufficient to fully establish the stored charge associated with the specified I_F; it is suggested to be between 10 and 30 microseconds. The reverse voltage (V_4) to the DUT is applied through S_2 at a rate determined by the rise time, t_2 and magnitude of the input pulse, V_2, which is applied slightly before the end of t_1. The negative supply, V_4, is to be specified. It should be sufficiently low to prevent the peak voltage, $V_{RM(REC)}$, from exceeding the rated reverse voltage, V_{RWM} , of the device under test. For most repeatable measurement, 30 V is recommended. The point of contact on the leads shall be no less than .375 inch (9.52 mm) from the body.

4.5.6 Peak reverse power characterization. The testing to characterize P_{RM} shall be conducted in accordance with method 4065 of MIL-STD-750. The following details shall apply:

a. During the test, the voltage (V_{BR}) shall be monitored to verify it has not collapsed.

b. Any collapse in V_{BR} during or after the test, or rise in leakage current (I_R) after the test that exceeds I_{R1} in table I, shall be considered a failure to that level of applied P_{RM}.

c. Progressively higher levels of P_{RM} shall be applied until failure occurs on all devices within the chosen sample size to characterize each sublot.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance	3101	See 4.3.1	$Z_{\theta JX}$			°C/W
Forward voltage	4011	Condition B, duty cycle ≤ 2 percent (pulsed) (see 4.5.1) tp = 8.3 ms (maximum)	V_{FM1}			
1N6626, US		I_{FM} = 4.0 A dc			1.5	V dc
1N6627, US		I_{FM} = 4.0 A dc			1.5	V dc
1N6628, US		I_{FM} = 4.0 A dc			1.5	V dc
1N6629, US		I_{FM} = 3.0 A dc			1.7	V dc
1N6630, US		I_{FM} = 3.0 A dc			1.7	V dc
1N6631, US		I_{FM} = 2.0 A dc			1.95	V dc
Forward voltage	4011	Condition B, duty cycle ≤ 2 percent (pulsed) (see 4.5.1) tp = 8.3 ms (maximum)	V_{FM2}			
1N6626, US		I_{FM} = 2.0 A dc			1.35	V dc
1N6627, US		I_{FM} = 2.0 A dc			1.35	V dc
1N6628, US		I_{FM} = 2.0 A dc			1.35	V dc
1N6629, US		I_{FM} = 1.4 A dc			1.40	V dc
1N6630, US		I_{FM} = 1.4 A dc			1.40	V dc
1N6631, US		I_{FM} = 1.4 A dc			1.60	V dc
Reverse current leakage	4016	Condition A or B, V_R = rated V_{RWM} (col. 2 of 1.3.2); pulsed (see 4.5.1)	I_{R1}		Col. 2 of 1.4	µA
Breakdown voltage	4022	I_R = 50 µA dc pulsed (see 4.5.1)	$V_{(BR1)}$	110 percent of col. 2 of 1.3.2		

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3		T_A = +150°C				
High temperature operation						
Reverse current leakage	4016	Condition A or B, pulsed (see 4.5.1), V_R = rated V_{RWM} (col. 2 of 1.3.2)	I_{R2}		Col. 3 of 1.4	µA dc
Low temperature operation:		T_A = −65°C				
Forward voltage	4011	Condition B, I_{FM3} = 50 percent of I_{F1} (col. 6 of 1.4); pulsed (see 4.5.1)	V_{FM3}		110 percent of col. 6 of 1.4	V
Breakdown voltage	4022	I_R = 50 µA dc, pulsed (see 4.5.1)	$V_{(BR2)}$	Col. 2 of 1.3.2		V
Subgroup 4						
Forward recovery voltage	4026	I_F =1 A; t_r = 12 ns	V_{FRM}			V
1N6626, US					8	
1N6627, US					8	
1N6628, US					8	
1N6629, US					12	
1N6630, US					12	
1N6631, US					20	
Capacitance	4001	V_R = 10 V dc, f = 0.1 to 1 MHz	C_T		40	pF
Low current reverse recovery time	4031	Condition B; see 4.5.5.1 and figure 5 herein	t_{rr1}			ns
1N6626, US					30	
1N6627, US					30	
1N6628, US					30	
1N6629, US					50	
1N6630, US					50	
1N6631, US					60	

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 − Continued						
High current reverse recovery time	4031	Condition D, see 4.5.5.2 and figure 6 herein; I_F = 2 A, di/dt = 100 A/µs.	t_{rr2}			ns
1N6626, US					45	
1N6627, US					45	
1N6628, US					45	
1N6629, US					60	
1N6630, US					60	
1N6631, US					80	
Peak recovery current			$I_{RM(REC)}$			A
1N6626, US					3.5	
1N6627, US					3.5	
1N6628, US					3.5	
1N6629, US					4.2	
1N6630, US					4.2	
1N6631, US					5.0	
Scope display evaluation	4023	Condition A, stable only (see 4.5.4)				
Subgroup 5						
Not applicable						
Subgroup 6						
Forward surge	4066	Condition A, I_{FSM} = rated I_{FSM} (col. 5 of 1.3.2) ten surges of 8.3 ms each at 1 minute intervals. I_O = 0, V_{RWM} = 0, T_A = +25°C				
Electrical measurement		See table I, subgroup 2.				
Subgroup 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Group E inspection (all quality levels) for qualification and requalification only.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			45 devices, c = 0
Thermal shock (glass strain)	1056	20 cycles, condition D except low temperature shall be achieved using liquid nitrogen (−195°C) Visual for cracked glass.	
Temperature cycling	1051	500 cycles, −65°C to +175°C.	
Hermetic seal	1071	Gross leak.	
Electrical measurement		See table I, subgroup 2 and table III, steps 1 and 2.	
Subgroup 2			45 devices, c = 0
Blocking life	1048	T_A = +150°C; t = 1,000 hours +65, −0 hours; dc = 80 to 85 percent rated V_R.	
Electrical measurement		See table I, subgroup 2.	
Subgroup 4			N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 5			22 devices, c = 0
Barometric pressure (reduced) 1/	1001	1N6626 through 1N6628 at 8 mm Hg. 1N6629 through 1N6631 at 33 mm Hg.	
Subgroup 8 2/			
Peak reverse power	4065	P_{RM} shall be characterized by the supplier and this data shall be available to the Government. Test shall be performed on each sublot.	
Electrical measurement		See 4.5.6.	

See footnotes at end of table.

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TABLE II. Group E inspection (all quality levels) for qualification and requalification only - Continued.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 9			45 devices
Resistance to glass cracking 1/	1057	Test condition B, step stress to destruction by increasing cycles or up to a maximum of 25 cycles.	
Subgroup 10			22 devices, c = 0
Forward surge	4066	Condition A, I_{FSM} = rated I_{FSM} (see 1.3.2, col. 5); ten surges of 8.3 ms each at 1 minute intervals, I_O = 0, V_{RWM} = 0.	
Electrical measurement		See table I, subgroup 2.	

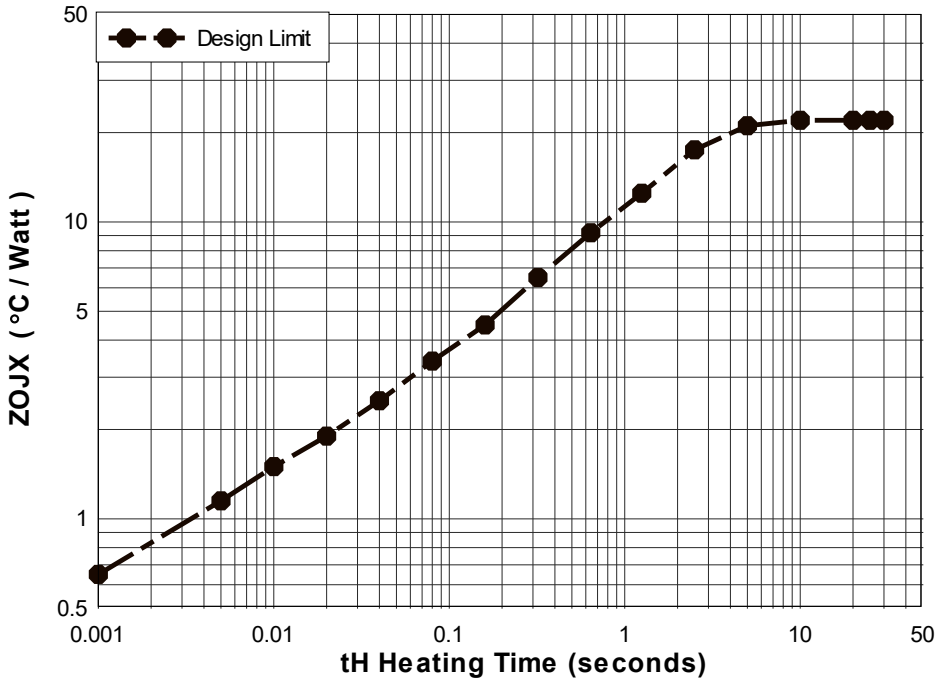
1/ Also applies to US suffix versions.
2/ The sample size for this step stress requirement shall be determined by the supplier. A statistically significant sample size is required.

TABLE III. Groups B and E delta measurements. 1/ 2/ 3/ 4/ 5/ 6/

Step	Inspection	MIL-STD-750		Symbol	Limits 4/		Unit
		Method	Conditions		Min	Max	
1.	Forward voltage change	4011	Condition B, I_{FM} = col. 5 of 1.4, pulsed (see 4.5.1)	$\Delta V_{FM}1$	±50 mV change from previous measured value		
2.	Reverse current change 1N6626 through 1N6631	4016	Condition A or B; V_{RM} = col. 2 of 1.3,	ΔI_{R1}		+2	µA

1/ The delta measurements for group B, table E-VIA (JANS) of MIL-PRF-19500 are as follows: Subgroup 5, see table III herein, steps 1 and 2.
2/ The delta electrical measurements for group B, table E-VIB (JAN, JANTX and JANTXV) of MIL-PRF-19500 are as follows: subgroup 2, see table III herein, steps 1 and 2.
3/ The delta measurements for group E, table E-IX of MIL-PRF-19500 are as follows: Subgroup 1, see table III herein, steps 1 and 2.
4/ See 1.4 herein.
5/ Devices which exceed table I, subgroup 2 (group A) limits for this test shall not be accepted.
6/ Also applies to US suffix versions.

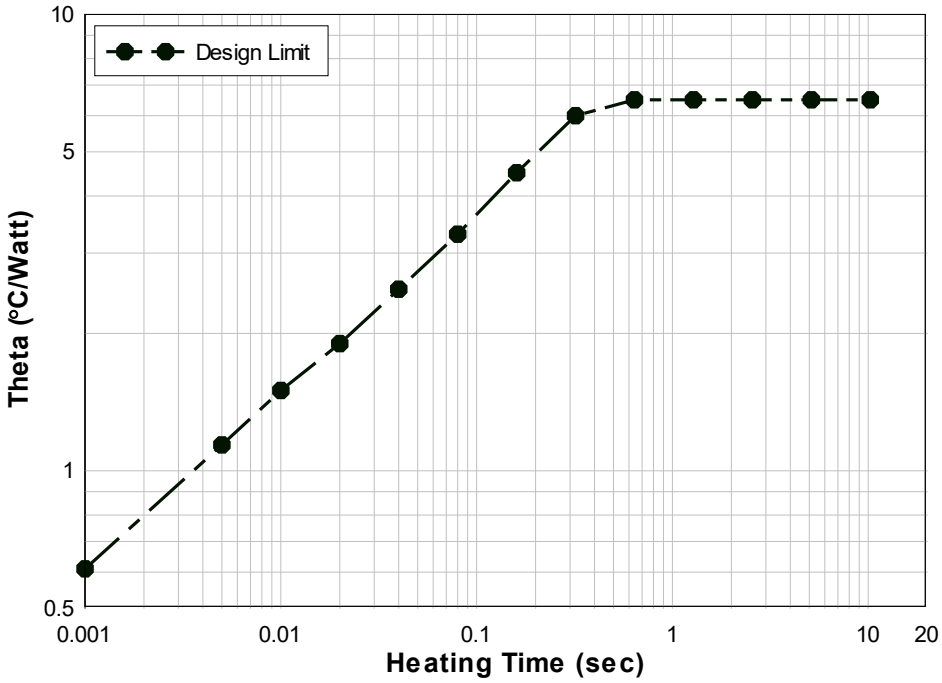
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$Z_{\theta X}$ = 1.5°C/W at 10 ms
$R_{\theta JEC}$ = 22°C/W

FIGURE 3. Thermal impedance curve for axial leaded package.

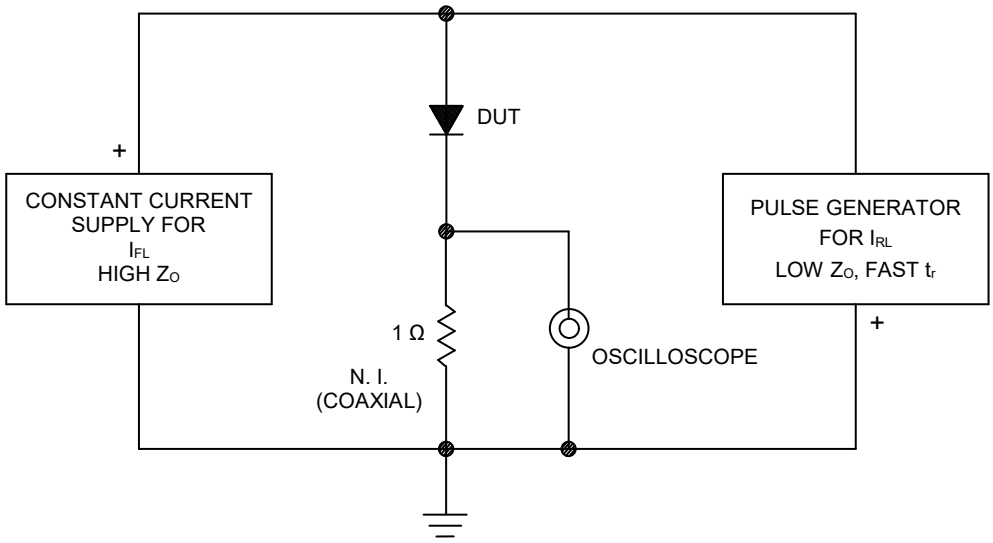
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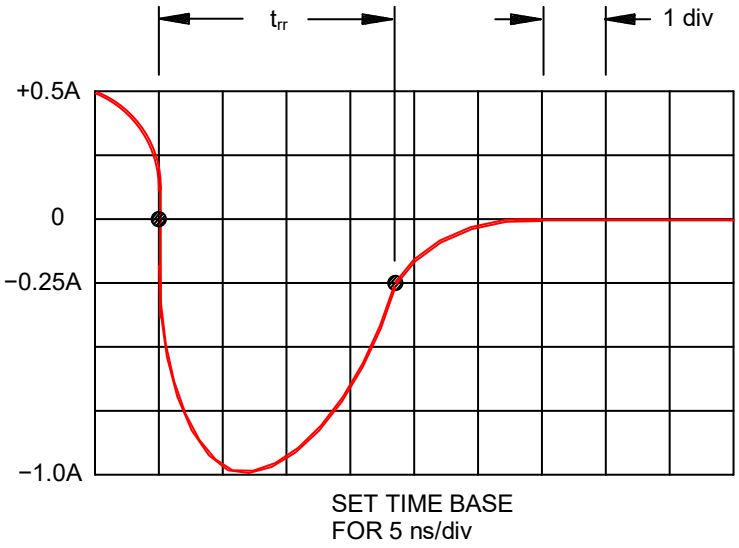


$Z_{\theta X}$ = 1.5°C/W at 10 ms
$R_{\theta JEC}$ = 6.5°C/W

FIGURE 4. Thermal impedance curve for surface mount package.

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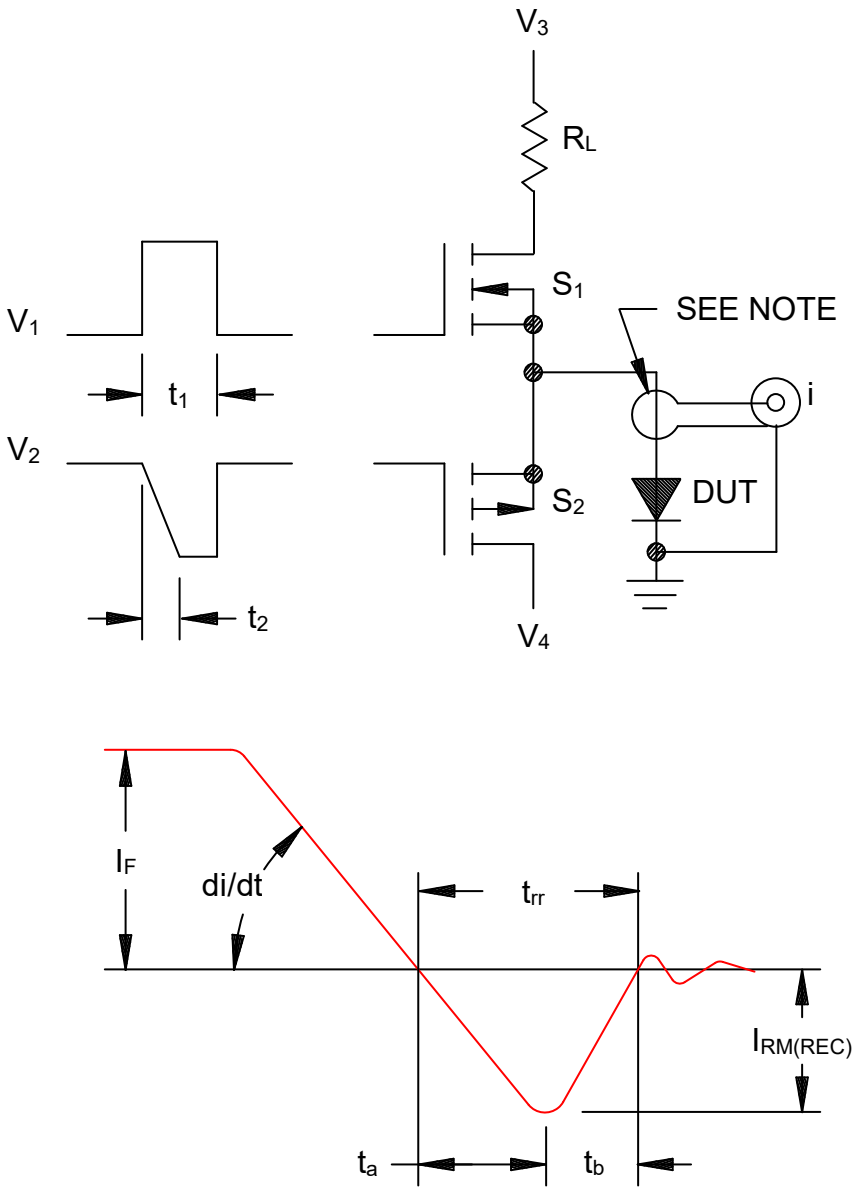




NOTES:
1. Oscilloscope: Rise time ≤ 3 ns; input impedance = 1 MΩ; 22 pF.
2. Pulse generator: Rise time ≤ 8 ns; source impedance = 10 Ω.

FIGURE 5. Low current reverse recovery time test circuit and characteristic waveform.

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NOTE: Current sensing transformer; alternatively a low inductance resistor may be used, from the cathode to ground, for current vs time measurements.

FIGURE 6. High current reverse recovery test circuit and waveform.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

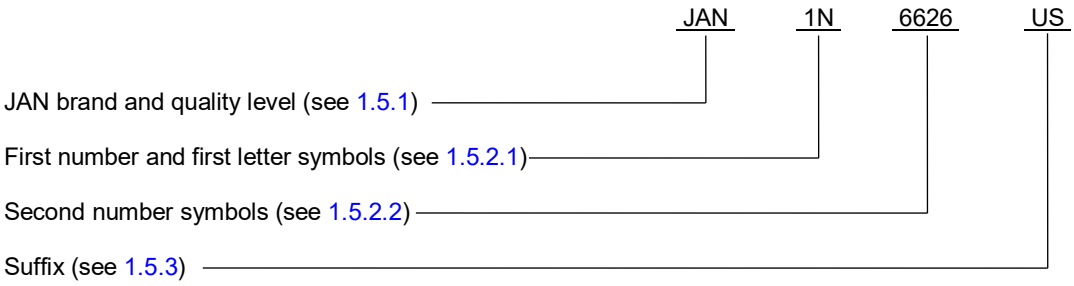
b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.5.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 PIN construction example. The PINs for encapsulated devices are constructed using the following form.



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6.5 List of PINs. The following is a list of possible PINs available on this specification sheet.

PINs for devices of the base quality level	PINs for devices of the "TX" quality level	PINs for devices of the "TXV" quality level	PINs for devices of the "S" quality levels
Axial through-hole mount (similar to DO-204AL)			
JAN1N6626	JANTX1N6626	JANTXV1N6626	JANS1N6626
JAN1N6627	JANTX1N6627	JANTXV1N6627	JANS1N6627
JAN1N6628	JANTX1N6628	JANTXV1N6628	JANS1N6628
JAN1N6629	JANTX1N6629	JANTXV1N6629	JANS1N6629
JAN1N6630	JANTX1N6630	JANTXV1N6630	JANS1N6630
JAN1N6631	JANTX1N6631	JANTXV1N6631	JANS1N6631
Surface mount version (US)			
JAN1N6626US	JANTX1N6626US	JANTXV1N6626US	JANS1N6626US
JAN1N6627US	JANTX1N6627US	JANTXV1N6627US	JANS1N6627US
JAN1N6628US	JANTX1N6628US	JANTXV1N6628US	JANS1N6628US
JAN1N6629US	JANTX1N6629US	JANTXV1N6629US	JANS1N6629US
JAN1N6630US	JANTX1N6630US	JANTXV1N6630US	JANS1N6630US
JAN1N6631US	JANTX1N6631US	JANTXV1N6631US	JANS1N6631US

6.6 Substitution and supersession information.

6.6.1 Inactive for new design. The surface mount device types using the "U" suffix covered by previous revisions of this document are inactive for new design as of the date of this revision (15 March 2020).

6.6.2 Superseded PINs. For PIN purposes, the "U" suffix has been superseded by "US" suffix. The non-definitive "U" suffix that was in previous revision of this specification sheet has been removed from this revision. The definitive "US" suffix should be used for future device identification, marking and procurement.

6.6.3 Substitutability of devices. The "US" suffix devices are form, fit, and functionally interchangeable with the superseded "U" suffix devices.

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6.7 Application data.

6.7.1 Half-sine-wave application for the worst-case (highest voltage) 1N6631. For a PCB mounting with FR4 material where the full 1.4 Amp I_O rating (half-sine wave) is used at a T_J of +150°C and ambient temperature of 25°C as shown on the first page ratings, the following steps guide the user in what the PCB pad size will need to be with 1 oz, 2 oz, and 3 oz copper. For axial-leaded, the lead length for mounting will be 0.187 inch (4.76 mm) or less from body to entry point on PCB surface.

a. Use the I_O versus Po curve on figure 7 to look up 1.4 amp (X-axis) and follow up to T_J = +150°C between the two lower curves for 2.48 watts.

b. Calculate maximum thermal resistance needed (150ºC to 25ºC) / 2.48 W = 50ºC/W.

c. Look up thermal resistance of 50ºC/W on Y-axis using a thermal resistance versus pad area plot on one of the three curves on figure 8 for different weights of copper cladding and then intersect curve horizontally to get answer. These curves assume still air, horizontal position.

d. In this example, the answer is: 1 oz PCB = 0.6 in^2 (387 mm^2), 2 oz PCB = 0.32 in^2 (206.5 mm^2), 3 oz PCB = 0.21 in^2 (134.5 mm^2) for each pad.

e. Add a conservative guard-band to the pad size (larger) to keep T_J below +150°C.

6.7.2 Square-wave application with 1N6626 to 1N6631. For a PCB mounting example with FR4 material to support a 0.7 amp I_O square-wave switching at a 0.50 duty factor (50 percent duty cycle) at T_J = +125°C and ambient temperature of 55°C, the following steps guide the user in what the PCB pad size will need to be with 1 oz, 2 oz, and 3 oz copper.

a. Find size of copper pads on standard FR4 PCB to support operation at 0.7 amp I_O square wave switching at a 0.50 duty factor (50 percent duty cycle) at T_J = 125°C with T_A = 55°C.

b. Calculate peak I_F = 0.7A / 0.50 duty factor = 1.4 amps.

c. Use the V_F versus I_F curve in figure 9 to look up I_F = 1.4 A (Y-axis) and follow across to the T_J = 125°C curve for V_F = 1.42 V.

d. Calculate power = I_F x V_F x duty factor = 1.4 x 1.42 x 0.50 = 0.994 watts.

e. Calculate maximum thermal resistance needed (+125°C to 55°C) / 0.994 W = 70°C/W.

f. Look up thermal resistance of 70°C/W on the Y-axis using a thermal resistance versus pad area plot on one of the three curves on figure 8 for different weights of copper cladding and then intersect curve horizontally to get answer. Curves assume still air, horizontal position.

g. In this example, the answer is: 1oz PCB = 0.20 in^2 (129 mm^2), 2oz PCB = 0.11 in^2 (70.9 mm^2), 3oz PCB = 0.08 in^2 (51.6 mm^2) for each pad.

h. A conservative pad guard-band is optional since T_J is only +125°C. NOTE: Multilayer PCBs, forced air cooling will improve performance. Closed confinement of the PCB will do the opposite. Please use sound thermal management.

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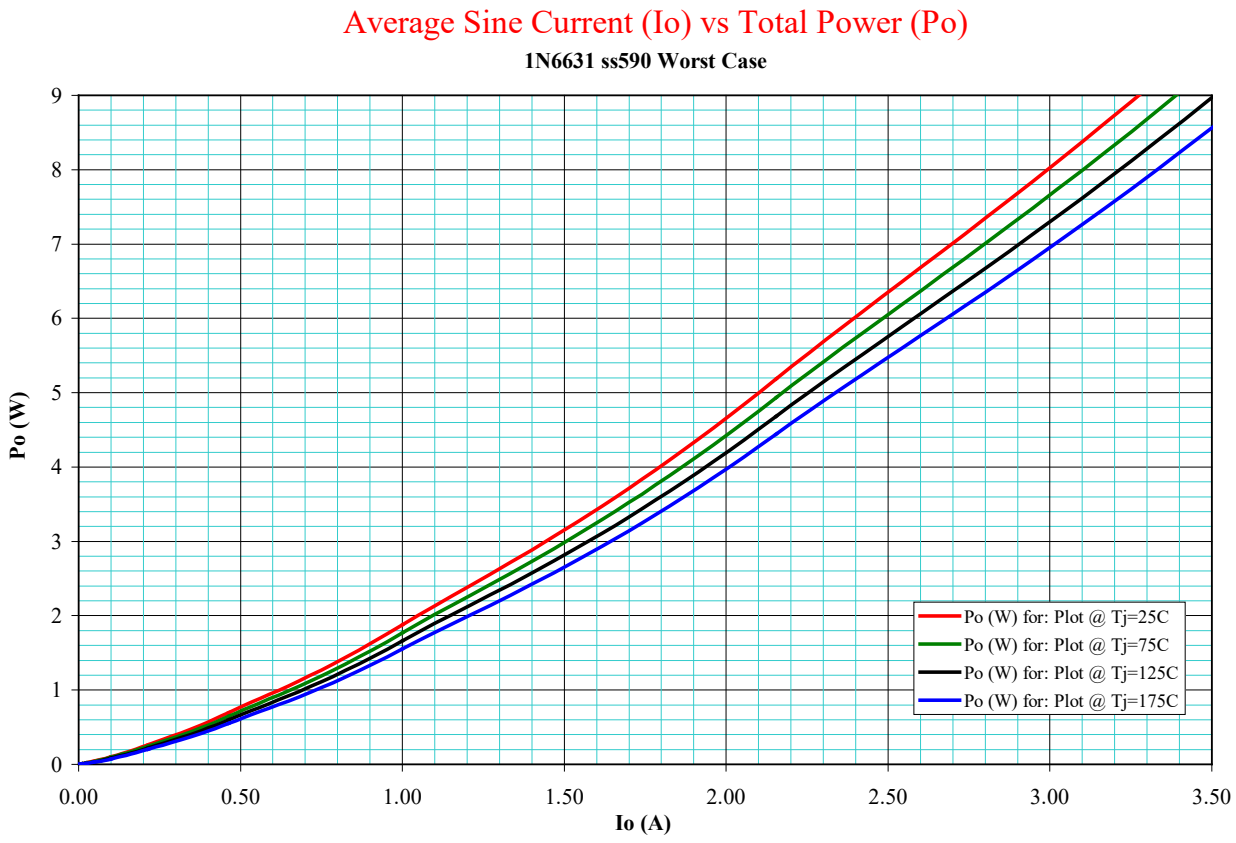


FIGURE 7. Rectifier power versus I_O (Average forward current for 1N6626 through 1N6631.

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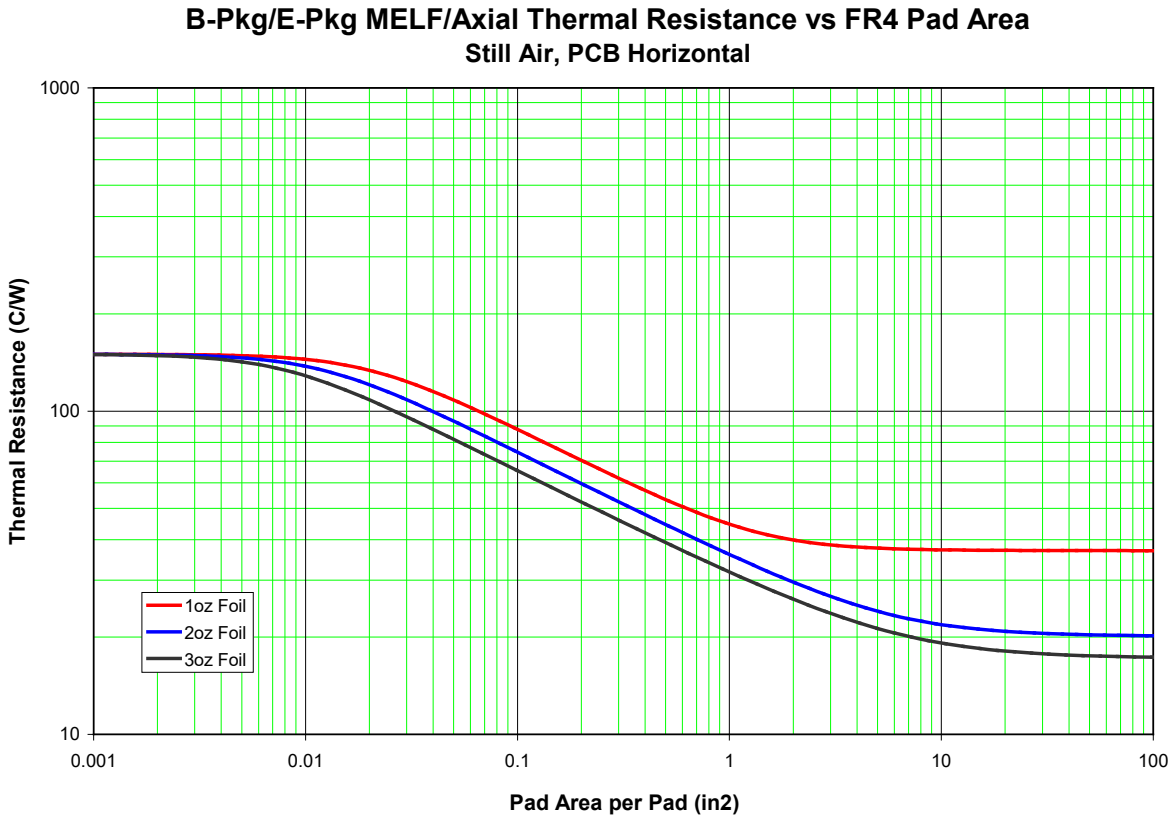


FIGURE 8. Thermal resistance versus pad area (for each pad) with 1, 2, and 3 oz Copper for 1N6626 through 1N6631.

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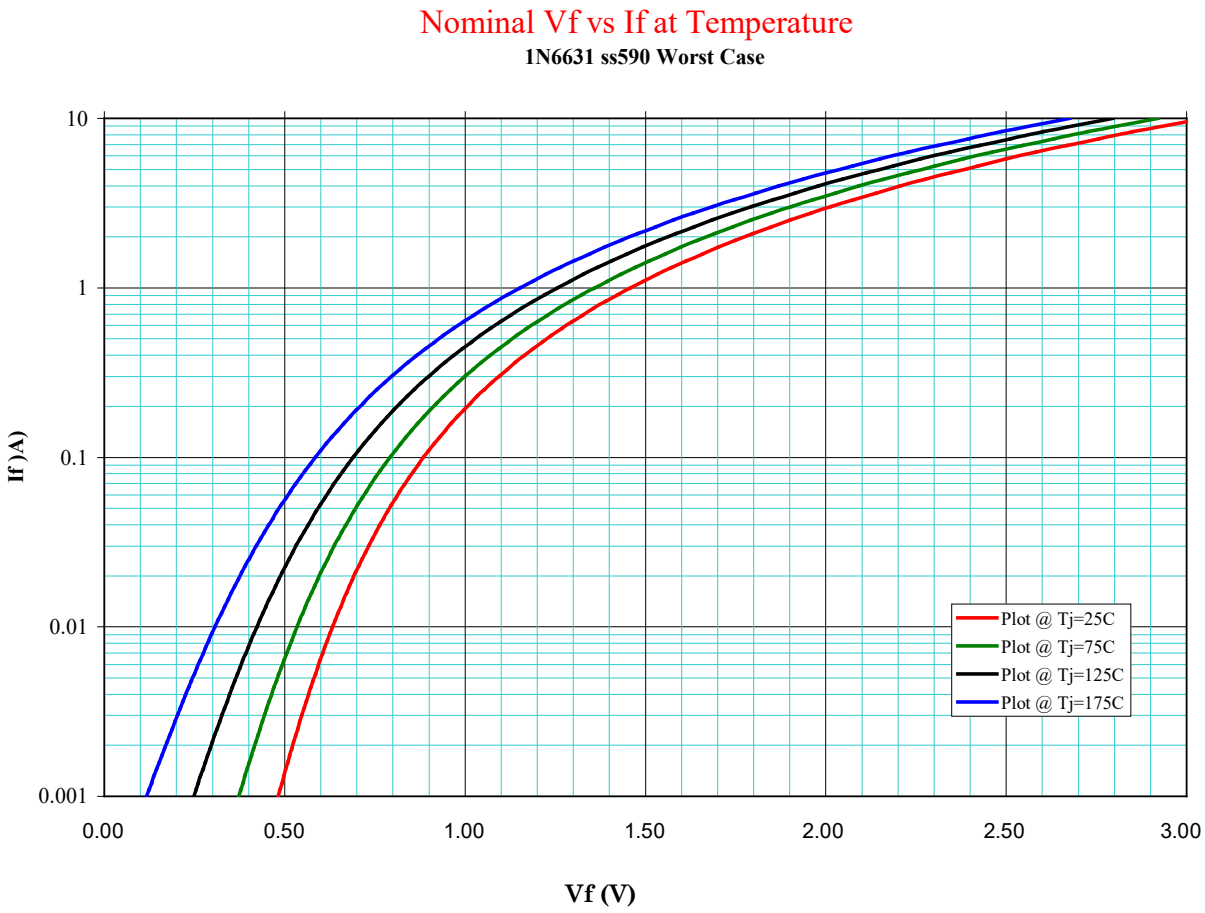


FIGURE 9. Forward voltage versus forward current for 1N6626 through 1N6631.

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6.8 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 693−1642 or DSN 850−6939.

6.9 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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Air Force − 85
NASA − NA
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